Anthony Ibarra

(Chicago, IL; Available to Start Onsite)

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Education:

University of Illinois Chicago (UIC) **Jan 2023 - May 2025**

Master of Science in Electrical and Computer Engineering Chicago, United States

Relevant Courses:

- Electromechanical Energy Conversion, Mechatronics Embedded Design

University of Illinois Chicago (UIC) **Aug 2017 - May 2022**

Bachelor of Science in Computer Engineering Chicago, United States

- Control Engineering

Languages:

- English, Spanish

Technical Skills:

Testing & Debugging:

- Soldering, Oscilloscope, Multimeter, benchtop powersupply, RF Analyzer (FieldFox)
- Unit Testing, Test-Driven Development (TDD), Pytest, uTest
- Hardware-in-the-Loop (HIL), Tracing, GDB Debugging

Tools & IDEs:

- VSCode, MATLAB, Simulink, Solidworks, Git, GitHub
- Altium Designer, WSL, Docker, Ubuntu, MySQL

Programming Languages:

- C, C++, Python, Bash, PLC-ladder logic(basics)

Experience:

Electronics Repair **present**

DIY

- Complete electronic repairs on multiple devices such as: Powertool battery pack repair, fixing lithium ion 12.0 AH with 20v/60v configurations. With the use of a multimeter, broken wires, bad solder joints, and dead or discharged battery cells were found accordingly and replaced or soldered respectively.
- Electrical controller repair where the smt pushbutton was broken or faulty discovered through testing and multimeter probing. Repair for 2 DC motors connection of broken wires and weak solder connection, fix through resoldering wires respectively.

Projects:

Autonomous-Driving Car – Mechatronics Embedded Design

University of Illinois Chicago (UIC) – Chicago, IL

- Led a multidisciplinary engineering team in the design and development of an autonomous embedded system, coordinating hardware, firmware, and testing milestones.
- Designed and implemented a motor driver circuit using MOSFET drivers (single FET, half-bridge, and full H-bridge configurations) controlled via PWM signals from a microcontroller for DC motor actuation.
- Developed a boost converter power supply to support multiple voltage rails for sensors, control electronics, and motor drivers.
- Implemented and tuned PID control algorithms in embedded firmware to regulate steering (servo motor) and vehicle velocity (DC motor) using encoder feedback.
- Designed and fabricated a custom PCB using Altium Designer, generating schematics, BOM, Gerber files, and manufacturing documentation for production.
- Assembled and debugged hardware including SMT and through-hole soldering, PCB masking, and wiring harness creation through crimping and soldering.
- Integrated line-tracking camera sensors and wheel encoders with the microcontroller, implementing sensor filtering and signal processing in embedded software.
- Performed hardware validation and debugging using oscilloscopes, multimeters, and benchtop power supplies to analyze PWM signals, power electronics behavior, and sensor outputs.
- Built a perfboard prototype backup system to ensure continued development and rapid hardware iteration during PCB testing.